BIO 304 . HUMAN ANATOMY & PHYSIOLOGY . WEEK 4 . MODULE 7 OF 16

Module 7: Nervous System

Topics: Neurons and Resting Membrane Potential, Action Potentials and Synaptic Transmission, CNS Organization: Brain and Spinal Cord, PNS and Autonomic Nervous System

TOPIC 1 OF 4 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

Neurons and Resting Membrane Potential

Neuron anatomy and how the resting potential is built and maintained.

The Science

Neuron anatomy first: dendrites receive, soma integrates, axon transmits, axon terminal releases neurotransmitter. The neuron is shaped for its job: a polarized communication device that can be a few millimeters long (interneuron) or over a meter long (motor neuron from spinal cord to foot).

Resting membrane potential is about -70 mV: the inside of the cell is negative relative to the outside. Two main forces keep it that way: the Na/K ATPase actively pumps 3 Na out and 2 K in (net export of positive charge, slightly negative contribution to the potential), and K leak channels let K flow out down its concentration gradient (leaving the inside more negative). The membrane at rest is mostly permeable to K; this is why the resting potential sits close to the K equilibrium potential of -90 mV but not exactly there (small contributions from Na and Cl).

Glia matter: astrocytes maintain the blood-brain barrier and clean up extracellular K and neurotransmitter. Oligodendrocytes wrap axons in myelin (CNS); Schwann cells do the same job in the PNS. Microglia are the immune cells of the brain. Ependymal cells line the ventricles and produce CSF. Glia outnumber neurons by roughly 10:1; the nervous system is mostly glia, not neurons.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a neuron with the parts labeled and then a graph of membrane potential over time at rest. They have to commit to the resting value.

Common misconceptions to address

- Students think the resting potential is set entirely by the Na/K pump. The pump is necessary to build the gradients, but the resting potential is set by ion permeability (mostly K leak).
- The minus sign of -70 mV is sometimes ignored. It is the entire point: the cell interior is negative.
- Glia are seen as bystanders. They are essential and outnumber neurons.

Order of operations (what to teach first)

Anatomy first, then the gradients (Na out, K in), then permeability and the resting potential, then glia as context.

Self-test prompts

- What is the typical resting membrane potential of a neuron?
- Why is the inside of the cell negative?
- Name the CNS glial cells and what each one does.

Why This Matters to Your Patients

Local anesthetics block voltage-gated Na channels and prevent action potentials in sensory neurons; the patient loses pain sensation. Cardiac glycosides (digoxin) inhibit the Na/K ATPase, which raises intracellular Na, which in turn raises intracellular Ca (through the Na/Ca exchanger), strengthening cardiac contraction. Multiple sclerosis (demyelination of CNS axons) is a glial disease. Hypokalemia and hyperkalemia both produce dangerous changes in membrane excitability, which is why potassium is one of the most carefully monitored electrolytes in the hospital. Every nurse needs to understand the relationship between potassium and the resting potential.

TOPIC 2 OF 4 . WEEK 4 . 12 CARDS (5 DOK 1 / 3 DOK 2 / 4 DOK 3)

Action Potentials and Synaptic Transmission

Phases of an action potential, propagation, and how chemical synapses pass the signal on.

The Science

The action potential is one of physiology's signature stories. Spend time on it; this is high-yield.

Phases: at rest, -70 mV. A stimulus depolarizes the membrane; if it reaches threshold (about -55 mV), voltage-gated Na channels open en masse, Na rushes in down its gradient, and the membrane swings positive (to about +30 mV). Then Na channels inactivate (a separate gate closes them) and voltage-gated K channels open; K rushes out, repolarizing the cell. K channels close slowly, producing a brief afterhyperpolarization. The Na/K pump resets the gradients (slowly, in the background). All-or-none: a suprathreshold stimulus always produces the same-size AP.

Propagation: unmyelinated axons propagate continuously, regenerating the AP at every patch of membrane (slow). Myelinated axons regenerate the AP only at nodes of Ranvier; the signal effectively jumps from node to node (saltatory conduction, much faster). Refractory periods (absolute, then relative) prevent backward propagation and limit firing frequency.

Synaptic transmission: AP reaches the terminal, opens voltage-gated Ca channels, Ca rushes in, vesicles fuse with the membrane, neurotransmitter is released into the synaptic cleft, binds receptors on the postsynaptic membrane, produces an EPSP (excitatory) or IPSP (inhibitory). Signal termination: reuptake, enzymatic breakdown, or diffusion away. EPSPs and IPSPs summate at the postsynaptic axon hillock; if the net depolarization reaches threshold, the postsynaptic cell fires.

Teaching This Topic

Before the video (drawing prompt)

Have students draw a labeled action potential trace from memory: time on x-axis, voltage on y-axis, with the phases marked. They have to commit before the video.

Common misconceptions to address

- Bigger stimulus equals bigger action potential. False: all-or-none means the AP is the same size regardless of stimulus strength. Bigger stimulus equals more frequent APs.
- K and Na are confused as to direction. Cue: Na is in for depolarization, K is out for repolarization.
- Saltatory conduction is sometimes described as 'jumping' which makes students think the AP literally jumps over membrane. It is regenerated at each node, just not at intervening membrane.

Order of operations (what to teach first)

Resting potential recap, then the AP phases (in order with ion movements), then all-or-none and refractory periods, then propagation, then synaptic transmission as the next-cell handoff.

Self-test prompts

- From memory, label the phases of an action potential on a voltage trace.
- Explain saltatory conduction in one sentence.
- What ion triggers neurotransmitter release at the axon terminal?

Why This Matters to Your Patients

Multiple sclerosis demyelinates CNS axons; saltatory conduction fails; symptoms depend on which tracts are affected. Local anesthetics block Na channels in sensory neurons selectively (because sensory axons are smaller and more easily blocked) and produce regional anesthesia. SSRIs raise synaptic serotonin by blocking reuptake; the antidepressant effect lags weeks behind the immediate pharmacology because downstream neuroplasticity changes are slow. Botulinum toxin blocks neurotransmitter release (presynaptic) and causes flaccid paralysis; tetanus toxin blocks inhibitory neurotransmitter release and causes rigid paralysis. Almost every neurology drug acts on some node of the AP-synapse cascade.

TOPIC 3 OF 4 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

CNS Organization: Brain and Spinal Cord

Brain regions and lobes, meninges, ventricles, and spinal cord anatomy.

The Science

The brain has four major regions to get right. Cerebrum: the cortex (gray matter) over white matter, divided into four lobes (frontal, parietal, temporal, occipital). Diencephalon: thalamus (sensory relay), hypothalamus (homeostasis HQ), epithalamus (pineal gland). Brainstem: midbrain, pons, medulla (vital reflexes, cranial nerves III-XII). Cerebellum: motor coordination and balance.

Functional cortex landmarks: precentral gyrus (frontal lobe) is the primary motor cortex; postcentral gyrus (parietal lobe) is the primary somatosensory cortex. Occipital lobe has the primary visual cortex; temporal lobe has auditory cortex and Wernicke area (language comprehension); Broca area is in the frontal lobe (language production). These landmarks are how you read a stroke.

Coverings and fluid: three meningeal layers (dura outside, arachnoid, pia next to the brain). CSF is made by choroid plexus in the ventricles, circulates through the subarachnoid space, and drains via arachnoid villi into the venous system. The spinal cord runs from the medulla to L1-L2. Below L1-L2, the cauda equina (loose bundle of spinal nerve roots) is what you stick a needle around for a lumbar puncture.

Teaching This Topic

Before the video (drawing prompt)

Have students draw the cerebrum, label the four lobes, and put a star on the precentral gyrus, postcentral gyrus, Broca area, and Wernicke area. They have to commit before the video.

Common misconceptions to address

- Students think the cerebellum makes movements happen. It modulates and coordinates movements; the motor cortex initiates them.
- The brainstem is often dismissed as small. It contains the vital reflex centers for breathing, heart rate, and consciousness.
- Broca and Wernicke get confused. Cue: Broca (frontal) is for production ('mouth in front'), Wernicke (temporal) is for comprehension ('ears on side').

Order of operations (what to teach first)

Four major regions, then cortex landmarks, then meninges and CSF, then spinal cord as the extension into the body.

Self-test prompts

- Name the four lobes of the cerebrum.
- Where is the primary motor cortex?
- What is the difference between Broca aphasia and Wernicke aphasia?

Why This Matters to Your Patients

Strokes are localized by neurological deficit. A patient with right-sided weakness, right-sided sensory loss, and Broca-type aphasia has had a left middle cerebral artery stroke. A patient with vertical gaze palsy, ipsilateral facial weakness, and contralateral body weakness has a brainstem lesion. Cerebellar strokes produce ataxia (uncoordinated movement) without paralysis. Increased intracranial pressure produces a stereotyped triad: hypertension, bradycardia, irregular breathing (Cushing triad). Nurses are often the first to spot neurological changes by serial assessment; knowing the anatomy means knowing where the problem is.

TOPIC 4 OF 4 . WEEK 4 . 6 CARDS (4 DOK 1 / 1 DOK 2 / 1 DOK 3)

PNS and Autonomic Nervous System

Cranial and spinal nerves, reflex arcs, and the sympathetic vs parasympathetic divisions.

The Science

PNS = cranial nerves and spinal nerves. Twelve pairs of cranial nerves (mostly head and neck; vagus, CN X, is the long-haul exception, innervating the heart, lungs, and much of the GI tract). Thirty-one pairs of spinal nerves, each with sensory (dorsal root) and motor (ventral root) fibers.

Somatic motor vs autonomic motor. Somatic is voluntary, single neuron from CNS to skeletal muscle, releases ACh at the NMJ. Autonomic is involuntary, two-neuron chain (pre-ganglionic from CNS, synapses in a ganglion, post-ganglionic to target), innervates smooth muscle, cardiac muscle, and glands.

Sympathetic ('fight or flight'): thoracolumbar origin (T1-L2). Short pre-ganglionic, long post-ganglionic. Most post-ganglionic releases norepinephrine at the target. Effects: pupils dilate, heart rate and force up, bronchi dilate, gut motility down, skeletal muscle vasodilation, skin vasoconstriction, glycogen breakdown. Designed for action.

Parasympathetic ('rest and digest'): craniosacral origin (CN III, VII, IX, X plus S2-S4). Long pre-ganglionic, short post-ganglionic. Releases ACh at target. Effects: pupils constrict, heart rate down, bronchi constrict, gut motility up, digestion active. Designed for maintenance.

Most organs receive both; the balance shifts depending on state. The reflex arc is the basic functional unit: receptor, sensory neuron, integration center (often spinal cord), motor neuron, effector.

Teaching This Topic

Before the video (drawing prompt)

Ask students to predict the sympathetic effect on each of: pupil, heart rate, gut, bronchi, glycogen. Then test the same list under parasympathetic activation.

Common misconceptions to address

- Sympathetic and parasympathetic are seen as opposing forces in a tug-of-war. More accurately, they have different baselines for different organs, and the balance shifts.

- Both divisions release ACh at some point. Cue: parasympathetic uses ACh at the target; sympathetic uses NE at the target (with ganglia using ACh in both cases).
- Students think autonomic effects are slow because they are involuntary. Sympathetic responses can be very fast (adrenaline surge in seconds).

Order of operations (what to teach first)

PNS overview (cranial and spinal nerves), then somatic vs autonomic distinction, then sympathetic and parasympathetic in detail, then the reflex arc.

Self-test prompts

- List the five components of a reflex arc.
- What does the sympathetic system do to the eye? The heart? The gut?
- Which cranial nerve has the broadest parasympathetic distribution?

Why This Matters to Your Patients

Beta-blockers blunt sympathetic effects on the heart (slower rate, less force, lower blood pressure). Atropine blocks parasympathetic effects at muscarinic receptors and is used to raise heart rate or dry secretions. Spinal cord injuries above T6 disrupt sympathetic outflow and produce autonomic dysreflexia: a noxious stimulus below the lesion (full bladder, pressure ulcer) triggers an unopposed sympathetic surge with severe hypertension and headache, while the parasympathetic system tries to compensate above the lesion. This is a true emergency. Diabetic autonomic neuropathy produces orthostatic hypotension, gastroparesis, and erectile dysfunction. Knowing the autonomic system is knowing why many medications produce predictable side effects.